Catering Management System - Systems Analysis & Design

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1. Stakeholders for the OnceForAll Catering Services System.

1. Customers

Customers are the main users of the system. They try to look for a suitable caterer for events like weddings, birthday parties, conferences, etc. The system assists customers by listing catering service providers in their vicinity within their budget and requirements like cuisine type and required services. Customers get quotes and read reviews from different caterers which they may find helpful. Since customers wish to make educational choices in a timely manner, the system must be user-friendly, reliable, and fast.

Relevance to the system:

- Fast and accurate listing of relevant caterers.
- Reading reviews and comparing quotes.
- Job request notifications and updates.
- Feedback and review of services offered by the caterers.

2. Caterers

Caterers are registered as businesses on the platform, and they provide catering services to clients. They depend on the OnceForAll system to find clients that need their services for events and parties. The system caters to Caterers by providing them with job opportunities according to their service preferences and geographical area. Caterers look for work through a profile which they can edit to reflect the services they offer, and they accept or reject requests from customers. They purchase credits to use for sending catering job quotes, and they earn a reputation based on customer reviews received.

Relevance to the system:

- Catering customer's job requests directly through the system.
- Editing business profile and services offered.
- Creating quotes with purchased credits.
- Building a good reputation through customer reviews.

3. System Administrators

The 'OnceForAll' Platform System Administrator's primary concern is servicing the Platform's operational continuity. Their main concern is to confirm the system is working fine and there are no security threats. Admins are responsible for user account administration (for both customers and caterers), caterer's verification, and payment processes. Moreover, they offer technical assistance and platform troubleshooting at their help desk.

System relevance:

- Steady state operation of the platform.
- Administration and support of user accounts.
- Verification of caterers' business information and legitimacy.
- System check-up and resolving any technical problems.

4. Owner/Manager (Edison Guacamole)

Currently the owner/manager of OnceForAll, Edison Guacamole is already interested in profits resulting from the platform. His responsibilities include business supervision covering business growth and performance. Edison captures the information available with the system like report on client's caterer's performance, report on the level of customer's satisfaction, and other financial figures like income from credits to guide strategic decisions in business.

System relevance:

- Obtaining reports regarding business and performance.
- Cross checking alignment of the system with business objectives.
- Taking a decision of service expansion or enhancement based on system data.

5. Payment Processors (e.g. PayPal)

Payment processors, such as PayPal, manage the transactions taking place in the system. Caterers use these outpayment systems to buy credits, while PayPal guarantees that the system is secure to use. Payment processors care about the efficient flow of payments and compliance with financial law.

Relevance to the system:

- Manage payment transaction security.
- Compliance with financial laws.
- Check correctness of credit purchases.

6. Potential Investors

Any business investor or stakeholder interested in OnceForAll as a prospecting business model becomes equally interested in the business. Their expectation is for the business to be profitable and for the market to be able to provide scale. Investors underline system's performance, growth and health to decide how far the business can go in terms of making profit and getting bigger.

Relevance to the system:

- Tracking the profits and growth ability for the system, platform, business.
- Analyzing business through system performance data.
- Taking actions depending on the upward growth of the system.

7. Oversight Entities

It is the responsibility of the governmental and regulatory authorities to make sure that OnceForAll complies with policies and legalities. There is emphasis on verifications such as business certifications and insurance compliance, caterers' verification, and data protection law compliance. They monitor the system for conformity to accepted practices and industry standards.

Relevance to the system

- Ensure that business and legal requirements are met.
- Auditing the practices regarding data protection and security.
- Confirmation of credentials and their validity.

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2. Functional and Non-Functional Requirements for the OnceForAll Catering Services System.

a) Functional Requirements

Functional requirements answer the What questions of system functionality. It defines the actions that a system must take in order to satisfy the user. These are the identified functional requirements for OnceForAll system:

1. **User Registration and Account Management**
   - **Description:** Customers and caterers need to create an account through an email or social media login. Customers need an account for submitting job requests, while caterers need to register accounts before they can provide services.
   - **Relevance:** Makes certain that all users are catered for the system and that they participate in transactions in a safe manner.

2. **Account Management**
   - **Description:** Users should be able to change personal information, change the password, change other preferences or even close the account.
   - **Relevance**: Help users manage their accounts and personal information.

3. **Caterer Profile Management**
   - **Description:** The caterers should be enabled to develop and update their profiles which includes business information and the services they provide, their geographical location, insurance covers, and client reviews.
   - **Relevance:** This helps caterers to market their services and provide essential information to customers.

4. **Submission of Job Requests**
   - **Description:** Customers should be capable of submitting job requests with clear specifications like event category, guest count, preferred cuisine, and required services.
   - **Relevance:** This is fundamental for the system to relate the appropriate caterers to the customer's event specifications.

5. **Caterer and Customer Matching Integration**
   - **Description:** The system should relate customer job orders to a database of caterers by their region, available services, and occupancy status.
   - **Relevance:** Guarantees that the system brings together customers and caterers in the most effective way.

6. **Filter & Search Options**
   - **Description:** Customers must be enabled to search and filter caterers by the type of cuisine offered, customer satisfaction ratings, price, and area where the caterer is based.
   - **Relevance:** Ensures that an instant response enhances the user's experience.

7. **Management and Generation of Quotes**
   - **Description:** Caterers have to have the ability to prepare and send forward bid documents for customer requests. They should contain information on likely expenses, the caterer's readiness to work on the request, and other pertinent information.
   - **Relevance:** This capability is important to augment negotiations and decision making between the customer and the caterer.

8. **Notification System**
   - **Description:** Both customers and caterers should receive either email or SMS notifications for updates regarding job requests, quotes, and progress changes.
   - **Relevance:** Communications that users deem pertinent are delivered, thereby making sure communication flows without hitches.

9. **Feedback and Rating System**
   - **Description:** Customers should be able to provide reviews, as well as rate the caterer after their event using some standardized metrics.
   - **Relevance:** While enable caterers to gain recognition, the feedback aspect helps in ensuring credibility and trust within the system.

10. **Closure of Job Request**
    - **Description:** After a customer chooses a caterer, the system should automatically close the job request and inform the non-selected caterers.
    - **Relevance:** This aids in streamlining the customer's contract expectations while simplifying the processes of closing catering business.

11. **Credit Payment**
    - **Description:** In order to submit quotes for catering jobs, caterers should be able to buy credits through an external payment mechanism like PayPal.
    - **Relevance:** It allows the system to function smoothly because the caterers will, through this method, be able to pay for the quote submission.

12. **Reporting & Analytics**
    - **Description:** The system must track the bookings made, the application's user satisfaction, and the income earned from credit purchases in order to generate reports on business indicators.
    - **Relevance**: Helps the owner/manager track business health and make data-driven decisions.

b) Non-Functional Requirements

The non-functional requirements describe the quality features of the OnceForAll system and are based on the FURPS+ taxonomy, which ensures the system is effective, safe and reliable. Following here is the Breakdown of non-functional requirements within the FURPS+ structure.

1. **Functionality (F)**
   - **Compliance**
     - **Description:** Regulation compliance includes the industry laws like data protection regulations (GDPR) and payment security laws (PCI DSS for payments).
     - **Relevance**: Makes sure that the platform is functional within legal boundaries and does not compromise users' sensitive information.

2. **Usability (U)**
   - **User-Friendly Interface**
     - **Description:** Interface for customers and caterers should be user friendly and easy to navigate and accessible.
     - **Relevance**: Better user experience, lower confusion, and ability for users to carry out tasks easily.

3. **Reliability (R)**
   - **Availability**
     - **Description:** The system has to be accessible at all times during the day and throughout the week, with only minimal downtime. Any maintenance work needs to happen when there is off peak demand and should be communicated to users beforehand.
     - **Relevance**: There is uninterrupted access for customers and caterers who need to post jobs and have transactions done.

   - **Fault Tolerance**
     - **Description:** The organization should be able to restore the system following a crash or failure promptly and without losing any important data. Automated backup systems should also be included.

- **Relevance**: It guarantees therefore no interruptions or severe damage to a transaction or job data will occur.

4. **Performance (P)**
   - **System Response Time**
     - **Description:** During peak times, all job requests, quotes, searches, and submissions should be completed with-in less than 2 seconds.
     - **Relevance:** This will deliver a smooth and responsive user experience.

   - **Scalability**
     - **Description:** The system ought to be able to support an increasing number of users and transactions from various locations within the country.
     - **Relevance:** This enables the business to be expanded nationally without the business suffering from system slow-downs or failures.

5. **Supportability (S)**
   - **Maintainability**
     - **Description:** Updating the system should be as effortless as changing documents because of modular systems and architecture with extensible code.
     - **Relevance:** Cost and time of updates like adding features should be simple.

   - **Extensibility**
     - **Description:** Adaptability of a platform should focus on adding features like payment systems and broader caterer services.
     - **Relevance:** Growth potential for the business and development flexibility is guaranteed.

Additional Conditions

- **Security (+)**
  - **Data Encryption**
    - **Description:** Cyber threats pose a significant risk, thus sensitive data such as user credentials and payment information must be routinely encrypted.
    - **Relevance:** Trust is increased because unauthorized access is mitigated.

  - **Access Control**
    - **Description:** User roles within the system such as admin, customer or caterer will govern system functionalities under role-based access control (RBAC).
    - **Relevance:** System functions that are sensitive are protected while being made available to authorized users only.

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3. Use Cases for the OnceForAll Catering Services System

a) Use Case List using the User Goal Technique

The **User Goal technique** involves identifying the primary goals that users aim to achieve when interacting with the system. Here is the list of use cases based on the User Goal technique:

Participating Actor	Use Case Name	Brief Use Case Description
Customer	Submit Catering Job Request	A customer submits a catering job request, detailing the event type, budget, number of people, service required, and additional instructions.
Customer	View Caterer Profiles	The customer views a list of caterers based on their submitted requirements and compares services, reviews, and quotes.
Caterer	Create Profile	A caterer creates and updates their business profile, providing essential details like services, location, and insurance information.
Caterer	Generate and Submit Quote	A caterer generates a quote based on a customer's job request and submits it for review, including the estimated cost and availability.
Customer	Accept Caterer's Quote	The customer reviews the caterer's quote and selects the caterer they wish to hire for their event.
Customer	Cancel Job Request	The customer cancels their job request before finalizing a caterer, either because they no longer need a service or have found a better option.
Caterer	Manage Credits	The caterer purchases and manages credits in order to generate quotes for customers.

Customer	Provide Feedback	After the event, the customer provides feedback and a rating for the caterer based on their experience.
System Administrator	Verify Caterer Details	The system administrator verifies the details of a caterer, including legal requirements like business name and insurance.
System Administrator	Generate System Reports	The administrator generates reports to analyze the system's performance, such as revenue from credits or customer satisfaction levels.

b) Additional Use Cases using the Event Decomposition Technique

The **Event Decomposition technique** helps to identify additional use cases that may be driven by temporal or state-based events. Below is the list of use cases identified through this technique:

Event	Type of Event	Use Case Name	Brief Use Case Description	Actor
Job Request Posted by Customer	Temporal	Match Caterers to Job Request	Once a customer posts a job request, the system automatically matches the request with suitable caterers based on location, type of event, and services required.	Customer
Caterer Submits Quote	Temporal	Notify Customer of New Quote	The system sends a notification to the customer when a caterer submits a quote for their job request.	Caterer
No Quotes Received in 48 Hours	Temporal	Notify Customer of Job Closure	If no caterers submit quotes within 48 hours, the system notifies the customer that their job request has been closed.	Customer
Caterer's Credits Expire	State	Notify Caterer to Purchase Credits	The system alerts the caterer when their credits are about to expire, prompting them to purchase new credits in order to continue submitting quotes.	Caterer
Customer Selects Caterer	State	Close Other Quotes	Once a customer selects a caterer, the system automatically notifies the other caterers that their quotes were unsuccessful.	Customer

Customer Leaves Review	State	Update Caterer's Rating	After the customer leaves a review, the system updates the caterer's average rating and displays the feedback on their profile.	Customer
Find Caterer	Temporal	Find Caterer	The system helps the customer find suitable caterers based on their job request details.	Customer
Print Credit Payments Report	Temporal	Print Credit Payments Report	The System Admin generates a report for Caterer's credit transactions, showing how much each has spent on credits.	System Admin
Provide Feedback	State	Provide Feedback	After the job is completed, the Customer provides a review for the Caterer. The system updates the rating based on this feedback.	Customer

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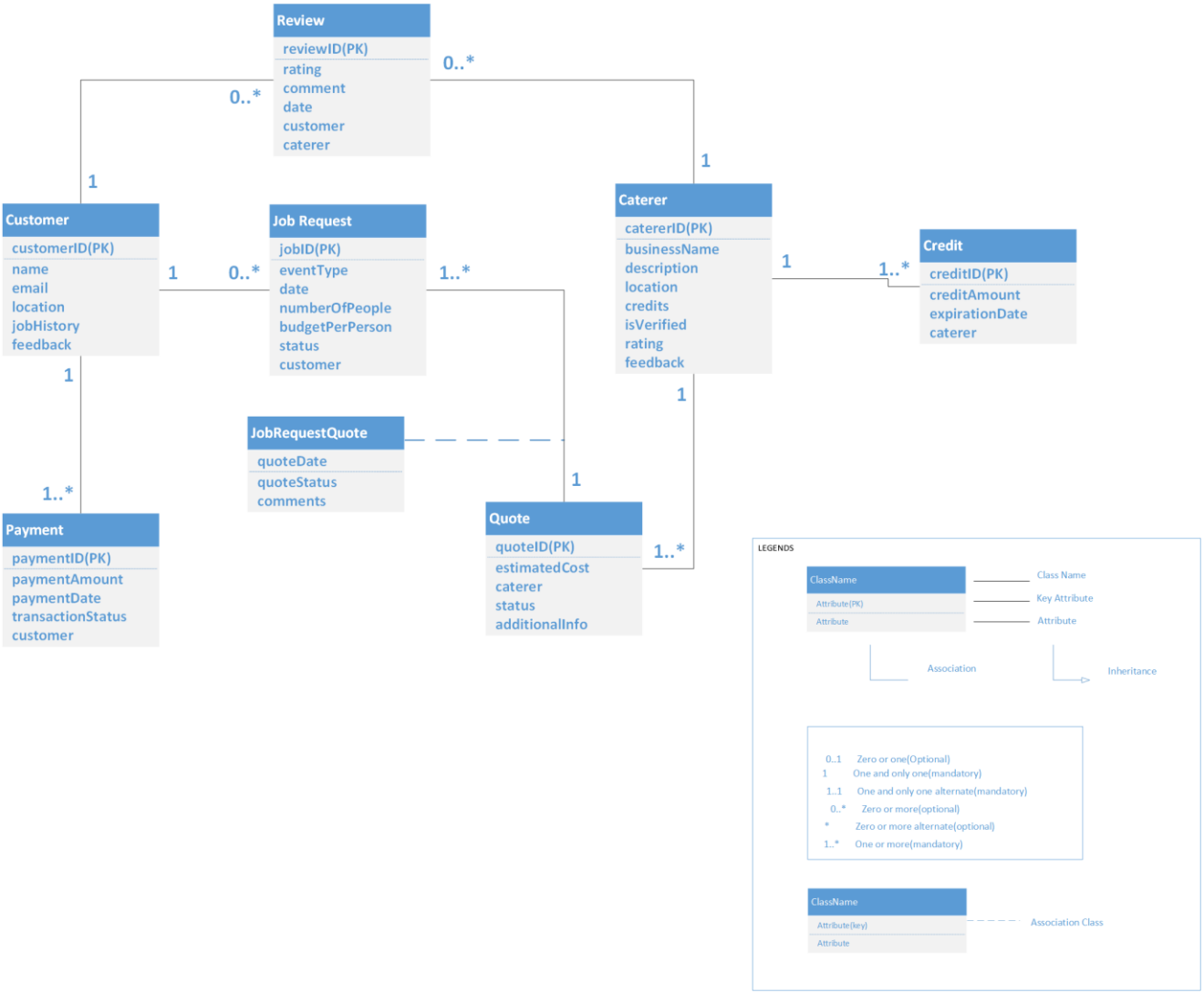
4. Domain Model Class Diagram for the OnceForAll System



Assumptions for Domain Model Class Diagram

a) **Customer and Job Request:** Customers may have zero or more job requests, within the limits of the system's configuration. Customers should be able to request as many requests as they like, or none at all.

b) **Job Request and Quote:** A minimum of one quote must be secured for each job request, although it is possible to have multiple quotes from different caterers. The system must be able to identify which quotes pertain to the job request and ensure that they are associated with the job request.

c) **Caterer and Quote:** A caterer may have many quotes, although every single quote is tied to just one caterer. The system has to be able to accommodate many quotes from caterers.

d) **Customer and Review:** A customer has the opportunity to give many reviews, or none at all. The system ought to know which customer it belongs to and how to manage these reviews.

e) **Caterer and Review:** A caterer may receive so many reviews from customers, or none whatsoever. The system should be able to modify the profile of the caterer based on the reviews given by customers.
f) **Caterer and Credit:** A caterer may qualify for so many credits through the submission of quotes. The system should be able to account properly and keep track of these credits to ascertain they are managed properly.
g) **Customer and Payment**: While customers don't make payments for credits directly, **caterers** do, so the system must track these external payments accurately.

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5. CRUD Matrix for the OnceForAll System

Use Case / Class	Customer	Caterer	Job Request	Quote	Review	Credit	Payment	JobRequestQuote
Match Caterers to Job Request	R	R	C	R				C
Notify Customer of New Quote		R	R	C				R
Notify Customer of Job Closure	R		R					D
Notify Caterer to Purchase Credits		R				R		
Close Other Quotes		R	R	D				D
Update Caterer's Rating	R				C			
Find Caterer	R	R	C					C
Print Credit Payments Report						R		
Provide Feedback	R				C			

LENGENDS:

- C: Create
- R: Read
- U: Update
- D: Delete

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6. Fully Developed Use Case Description for "Find Caterer"

Use Case Name:	**Find Caterer**
Scenario:	The Customer has to individually seek out various caterers that would fulfil their needs. The system must assist the users in locating the required caterers according to their location, event nature, preferred cuisine, and available services.
Triggering Event:	The customer describes and submits a job request detailing the event (what it is, when it will take place, where, how many guests will attend, budget, etc.) to the System.
Brief Description:	Once the Customer has entered their job request with specific requirements, the System will seek out a relevant list of Caterers which the Client can choose from. They can compare profiles as well as reviews before ultimately deciding on a suitable caterer.
Actors:	**Primary Actor:** Customer **Secondary Actor:** Caterer, System
Stakeholders:	**Customer:** Needs to find the best caterer that suits their event requirements. **Caterer:** Wants to be selected by Customers for catering jobs. **System Admin:** Maintains the system functionality to match Customers with Caterers efficiently.
Preconditions:	a) The Customer has to be logged into their account on OnceForAll. b) The Customer has to submit the Job Request along with its details (type of event, place, headcount, etc.). c) The Caterers should be registered and have their details saved in the system (services offered, area covered, reviews, etc.)
Postconditions:	a) The Customer is shown the list of Caterers that respond to the Customer's Job Request. b) A Job Request is created in the system associating the Customer with the possible Caterers. c) The Caterers are informed about the Customer's job request. d) The Caterers may view the job details, and provide bid responses.

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	Actor	System
Flow of Activities:	1. **Customer** submits their **Job Request** form, providing event details (type, location, number of people, etc.).	1. **System** receives the **Job Request** and processes it.
	2. **Customer** submits the form.	2. **System** checks the list of available **Caterers** who meet the **Customer's** requirements based on location, services, and availability.
	3. **Customer** waits for the system to process the matching.	3. **System** generates a list of **Caterers** that meet the **Customer's** criteria.
	4. **Customer** views the list of **Caterers** and compares their profiles, services, reviews, and quotes.	4. **System** displays the list of **Caterers** and their details, including reviews, services, and availability.

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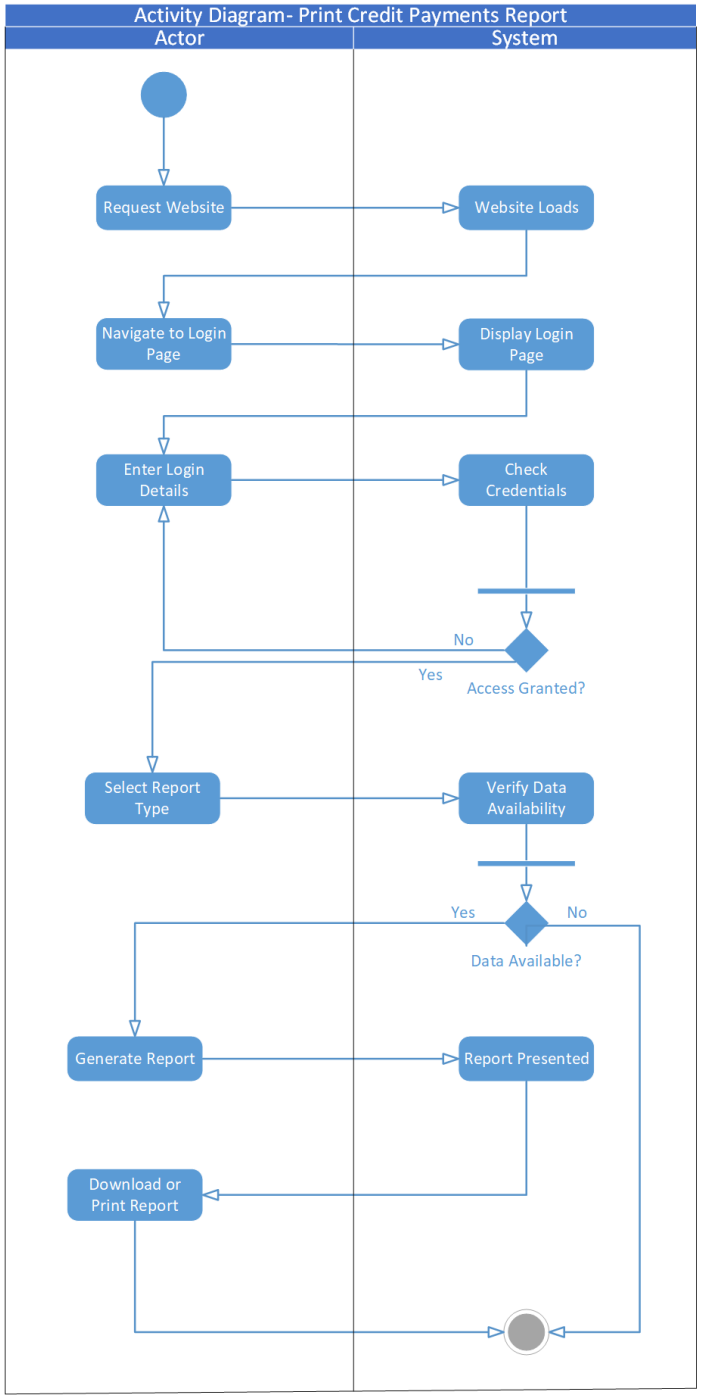
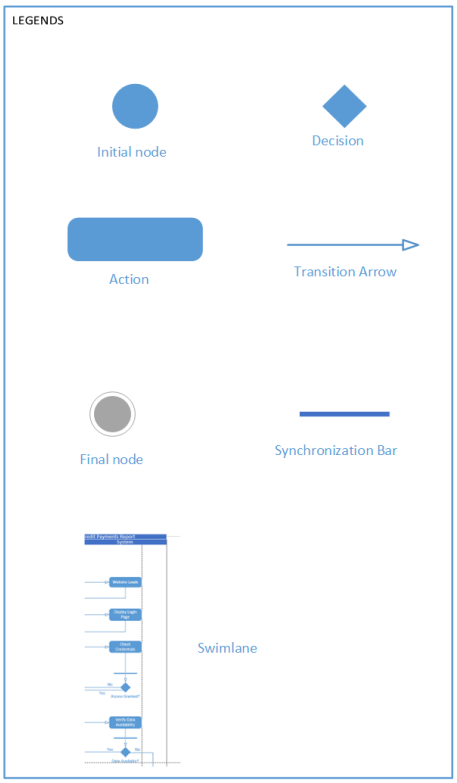
7. Activity Diagram for "Print Credit Payments Report"





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Activity Diagram Assumptions:

- **Login and Access:** To access the OnceForAll website, the customer must first request and log in. If the login is successful, they will have access to the system.
- **Report Selection and Data Verification:** After logging in, the customer will choose the Credit Payments Report. The system will determine whether the relevant data is accessible for report preparation.
- **Report Generation and Presentation:** The system generates and presents reports to customers based on available data. The customer can then choose whether to download or print the report.
- **Data Availability:** If data is lacking, the system will alert the consumer and end the procedure.
- **Flow Control:** The process will follow a simple flow with decisions based on login credentials and data availability. If any step fails (e.g., invalid credentials or missing data), the process will loop back or end.

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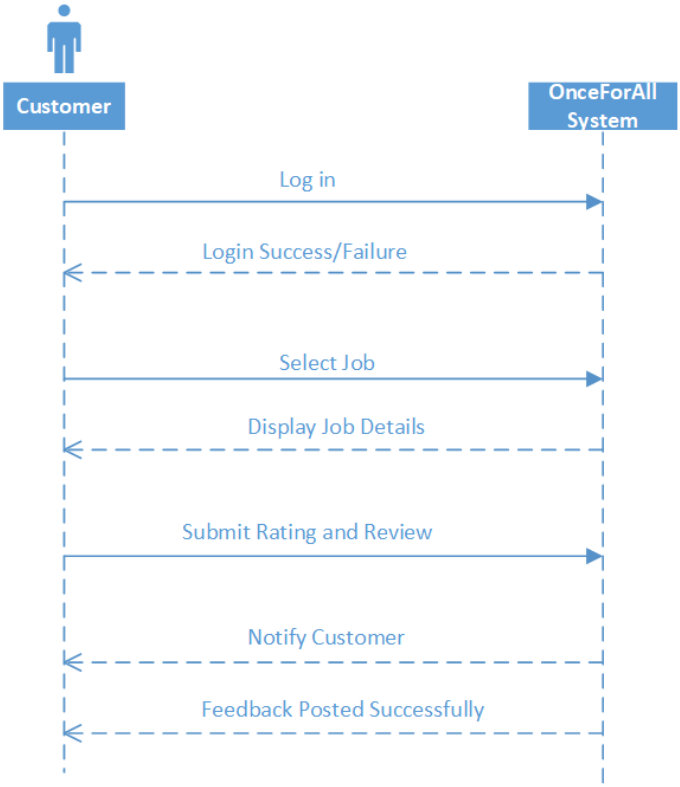
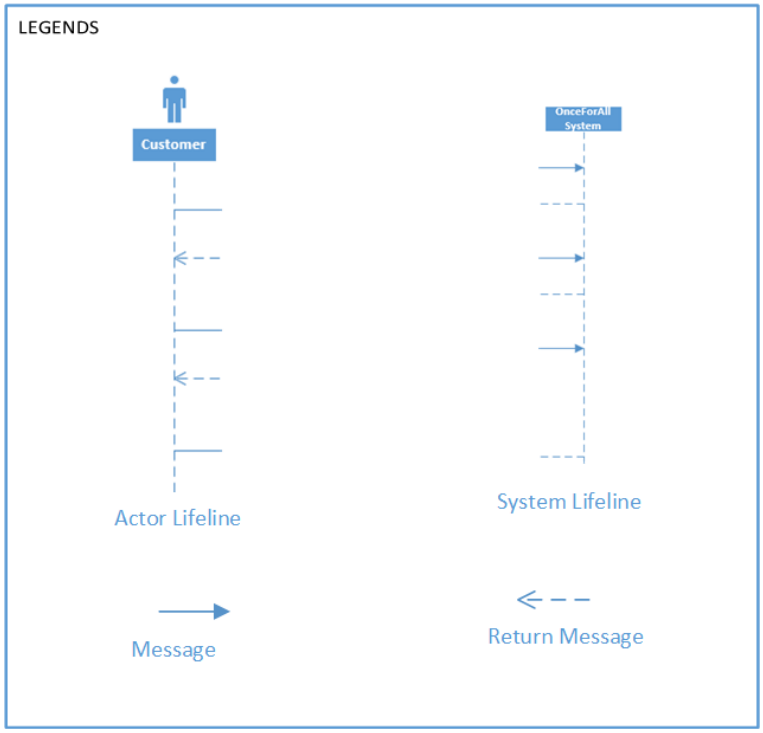
8. System Sequence Diagram for "Provide Feedback"





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Assumptions for System Sequence Diagram:

- **Login of the Customer:** The customer needs to login in the system. The system will verify the credentials and either allow access to the system or instruct the customer to enter the information again.
- **Job Selection for Feedback:** The Customer will choose a particular job that has been completed and requires feedback. The system fetches the job details to check if the job selected is the one intended.
- **Submission of Feedback:** The Customer will provide a score (usually out of five) and accompanying comments regarding the service. This feedback will be captured in the system together with any other comments and will also update the profile of Caterer with the new average score.
- **System Updates Caterer's Rating:** The system determines and refreshes the profile of the Caterer with the new average score based on the received comments.
- **Confirmation to Customer:** The system will notify the customer who posted comments, that their comments has been successfully published after confirming that feedback has been recorded in the system.
- **Flow Chart of Process**: The sequence follows a single line unless a certain condition goes wrong; in which case the sequence will rerun the process and return to Enter Login Details.

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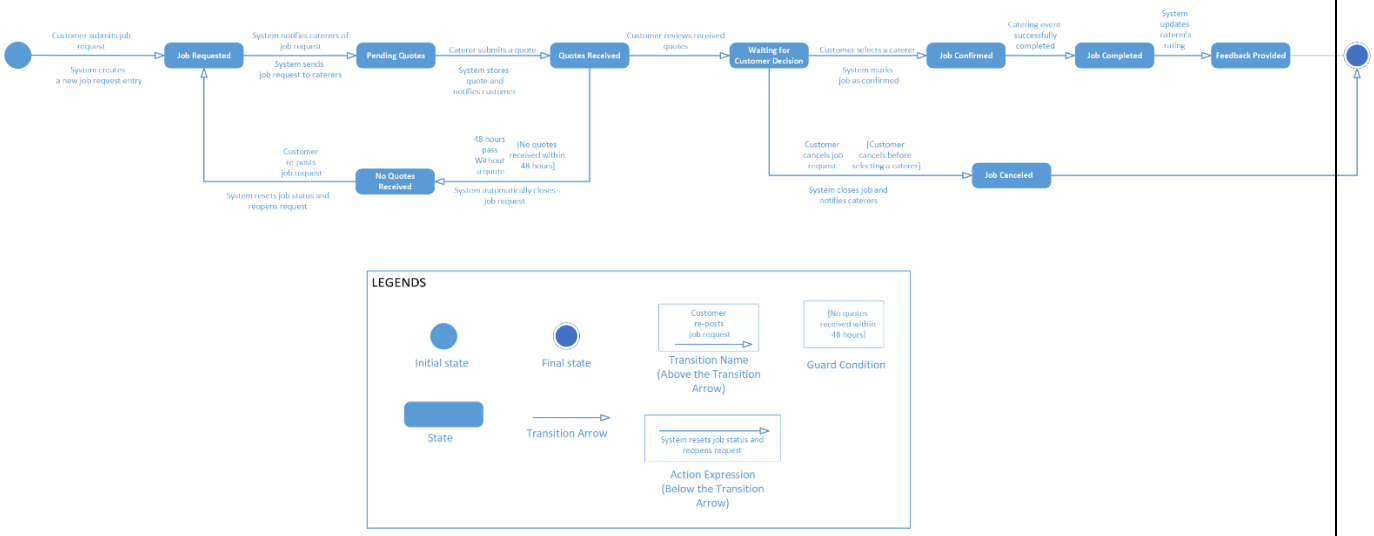
9. State Machine Diagram for the Catering Job Object



Assumptions for State Machine Diagram

- **Job Request Submission**: Submitting a job request creates a new entry in the system.
- **Pending Quotes:** The system notifies caterers to submit quotes for job requests.
- **Quotes Received**: Caterers can submit quotes, which the system stores and alerts the consumer.
- **Customer Decision:** The system confirms the selected caterer or cancels the job, notifying the caterers.
- **No Quotes Received**: If no quotes are received within 48 hours, the system will automatically close the task request.
- **Reposting Job Request**: If a customer reposts a job request, the system will reset the status.
- **Job Completion**: After the catering event, the system marks the work as finished.
- **Feedback Provided**: After the event, the **customer** provides feedback, and the system updates the **caterer's rating**.

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10. User Acceptance Test Plan for a Customer of the OnceForAll System

Use Case Name	Test Conditions	Expected Outcomes
Submit Catering Job Request	Customer submits job request with event type, number of people, and budget details.	A job order has been successfully submitted, and all the input information is preserved by the system.
View Caterer Profiles	A customer narrows the search for the caterers in accordance with the job request parameters (e.g. business location).	The system retrieves a list of appropriate caterers who can fulfill the requirements of the customer.
Accept Caterer's Quote	A customer chooses a caterer from the quoted caterers.	The system saves the selection made and sends an alert to the selected caterer.
Cancel Job Request	A customer withdraws from the job order prior to showing intent to accept a caterer's service.	The system changes the status of the job to 'Canceled' and alerts all the caterers that were involved.
Provide Feedback	A customer gives feedback on a completed event (score and description).	The customer feedback and score is reflected on the caterer's profile.
Notify Customer of New Quote	Caterer submits a quote in response to the customer's job request.	The customer is sent a notification which contains the quote from the caterer along with other relevant information.
Notify Customer of Job Closure	There were no accepted quotes by caterers for the job request within 48 hours of submission.	The customer gets informed of the closure of the job request due to no activity on it.
Close Other Quotes	A customer chooses a caterer out of a number of quoted caterers.	The other caterers receive the notification that the bids they submitted have been turned down.
Notify Caterer to Purchase Credits	The caterer attempts to submit a quote, but does not have enough credits.	The system communicates with the caterer to inform them that they need to buy new credits in order to keep proposing quotes.
Print Credit Payments Report	A system administrator prepares a report on the caterer's credit use.	A report of the credit purchases made by the caterers is provided through the system.